

SOPHIA BENZARI

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EDUCATION

B.S. Biomedical Engineering | University of Southern California

May 2028

Relevant coursework: Signal Processing, Data Analysis, Biomedical Instrumentation

Principles of Microeconomics & Physics (mechanics) | Harvard Secondary School Program

Summer 2023

Principles of Macroeconomics & Physics (electromagnetism) | Harvard Extension School

Summer 2025

RESEARCH EXPERIENCE

Undergraduate Researcher | Bionic Ear / Cochlear Implant Lab, USC

Sep 2025 – Present

- Designed and implemented a MATLAB-based multi-channel vocoder (8–16 channels) modeled on cochlear implant signal processing pipelines
- Integrated 3+ custom signal-processing modules into an existing research codebase to support active lab studies
- Assisted with human-subject research: survey design, qualitative interviews with participants, and data collection on auditory perception

Biomedical Engineering Co-Founder & Research Lead | InflAura – Digital Health Venture

Sep 2025 – Present

- Conducted 15+ customer discovery interviews with PCOS, thyroid, and reproductive-health users to identify unmet needs in wearable health monitoring
- Designed a single inflammation-risk indicator synthesizing 3 wearable biomarkers (HRV, resting heart rate, sleep) grounded in peer-reviewed inflammation research
- Executed hypothesis-driven validation plans testing 2 user segments and 3 pricing tiers with defined quantitative success metrics

CLINICAL EXPERIENCE

Clinical Intern | Clinique du 16 Novembre, Rabat, Morocco

Jun – Jul 2024

- Shadowed oncologists during 20+ patient consultations covering cancer diagnostics, treatment planning, and clinical decision-making
- Supported patient intake and routine clinical workflows for approximately 5 patients per day in a fast-paced hospital environment

Manufacturing & Quality Intern | Cosmetique Medical, Rabat, Morocco

Oct 2022 – Jan 2023

- Verified GMP-compliant production processes across formulation, production, and quality control stages for medical-grade cosmetic products
- Contributed to formulation development and quality control procedures across multiple production batches

LEADERSHIP & ADDITIONAL EXPERIENCE

Session Leader | SCout – Science Outreach at USC

Jan 2025 – Present

- Organized and led a team of student volunteers delivering weekly hands-on STEM lessons to K–3 classrooms; ranked in the top 3 performing groups program-wide
- Managed session logistics, scheduling, and participant communications end-to-end; led orientation briefings for new volunteers on safety and community standards

Lab Assistant & Woodshop Apprentice | Baum Family Makerspace, USC

2025 – Present

- Trained on all woodshop machines; certified to operate and supervise equipment in a multi-club collaborative workspace
- Primary wood-related resource for student clubs across the makerspace, supporting project planning, fabrication, and cross-disciplinary collaboration

Sales Associate | USC Bookstore

Sep 2025 – Jan 2026

- Managed 50+ customer interactions per shift in a high-traffic retail environment; handled transactions, inventory coordination, and team communication

SKILLS & LANGUAGES

Technical: MATLAB, Fusion 360, C++, Signal Processing, Python, Data Analysis, Excel

Languages: English (native), French (native), Arabic (conversational), Spanish (basic)